

TMS, DMS and CMS Usage Guide for Falcon BMS 4.38.1

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Chapter 1

CMS – Countermeasures Management Switch

1.1 Concept

The Countermeasures Management Switch (CMS) is a four-direction hat switch mounted on the side-stick controller. It controls the aircraft's defensive avionics:

- ALE-47 CMDS (Countermeasures Dispenser Set) — the chaff and flare dispensing system; and
- ECM (Electronic Countermeasures) — either external pods (ALQ-131/ALQ-184) or internal avionics (IDIAS) configurations, depending on block and variant.

CMS actuation procedures are presented in [Section 1.2](#) and block and variant differences in [Section 1.3](#).

For complete CMDS and ECM operation, see *TO 1F-16CMAM-34-1-1*, Dash-34, sections 2.7.1 and 2.7.2, respectively. Diagrams of the side-stick controller and throttle functions are provided in Dash-34 § 2.1.5. [Figure 1.1](#) below shows the CMS location on the F-16 side-stick controller.



Figure 1.1: F-16 Throttle and Side-Stick Controller. Image by Falconpedia (falcon4.wikidot.com), via Wikimedia Commons (https://commons.wikimedia.org/wiki/File:F-16_Side_Stick_Controller.jpg), licensed under the Creative Commons Attribution-Share Alike 3.0 Unported (CC BY-SA 3.0) license.

1.2 CMS Actuation

The tables below cover CMS actuation organized by system: CMDS (Manual, Automatic, and Semi-Automatic modes) and ECM (external pods ALQ-131/ALQ-184 and internal IDIAS).

All CMS actuation described in this chapter is *independent* of the Master Mode (NAV, A-A or A-G) selected.

Table columns are defined as follows:

- **Mode:** The operational context (Master Mode).
- **Direction:** CMS hat direction (Up, Down, Left, Right).
- **Action:** Press type (Short, Long, Long Hold).
- **Function:** What the CMS command activates or controls.
- **Effect / Nuance:** Resulting system behavior, including tactics and constraints.
- **Dash34:** Reference section in the Dash-34 manual.
- **Training:** Recommended BMS training mission(s) (see BMS Training Manual 4.38.1 for descriptions).

1.2.1 CMDS

The ALE-47 CMDS operates in three modes selected via the CMDS CCU MODE knob: Manual (MAN), Automatic (AUTO), and Semi-Automatic (SEMI). The single program to be executed in the chosen mode — Programs 1–4 — is selected via the PRGM knob on the CMDS panel.

Only Programs 1–4 are selectable. Programs 5 and 6 are independent of the PRGM knob position:

- Program 5 is activated via the cockpit slap switch — outside the scope of CMS — and is not covered in this guide; see Dash-34 § 2.7.2 for details.
- Program 6 is a special-purpose program always available for manual dispensing, independent of the PRGM knob position.

1.2.1.1 Manual Mode

In Manual (MAN) mode, the pilot runs countermeasure programs (1–4 and 6) via **CMS Up** and **CMS Left**.

Table 1.1: CMS Actuation with CMDS Manual Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS MAN	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once per press. No threat sensing; no automatic triggering. Overrides any AUTO dispensing, if running.	2.7.2.2	18, 28

Continued on next page

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS MAN	Left	Short	Execute Program 6	Program 6 can be dispensed independently of the PRGM knob position.	2.7.2.2	19

1.2.1.2 Automatic Mode

In Automatic (AUTO) mode, the ALE-47 CMDS dispenses the selected program (1–4) *continuously* in response to threats detected by the Radar Warning Receiver (RWR).

A **CMS Down** press grants consent for AUTO dispensing to start. It will continue until the pilot presses **CMS Right** or expendables are exhausted.

Even in AUTO mode, manual dispensing is still available.

Table 1.2: CMS Actuation with CMDS Automatic Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS AUTO	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once, interrupting any ongoing AUTO dispensing. After execution, AUTO resumes if the threat persists.	2.7.2.2	18, 28
CMDS AUTO	Left	Short	Execute Program 6	Program 6 is dispensed independently of the PRGM knob position. Overrides AUTO; after execution, AUTO resumes.	2.7.2.2	19
CMDS AUTO	Down	Short	Give Consent; Enable AUTO Dispensing	Grants consent for AUTO dispensing. RWR-detected threats trigger automatic execution of the program (1–4) selected via the CMDS panel PRGM knob. Dispensing continues until the threat clears, the pilot presses CMS Right, or expendables are exhausted. Consent state persists if the pilot switches to MAN mode. If the pilot re-engages AUTO, dispensing resumes immediately on the next detected threat.	2.7.2.1	18, 28
CMDS AUTO	Right	Short	Cancel Consent; Disable AUTO	Removes CMDS consent and places the ALE-47 in Standby. Automatic dispensing halts immediately. The pilot must re-issue CMS Down to resume AUTO operation.	2.7.2.1	18, 28

1.2.1.3 Semi-Automatic Mode

In Semi-Automatic (SEMI) mode, the CMDS control unit displays **DISPENSE RDY** and sounds a **COUNTER** voice message when a threat is detected. A **CMS Down** press grants consent and the ALE-47 CMDS then dispenses *one program*.

If the threat persists or a new threat appears, the CMDS issues **COUNTER** again.

Even in SEMI mode, manual dispensing is still available.

Table 1.3: CMS Actuation with CMDS Semi-Automatic Mode

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS SEMI	Up	Short	Execute Program 1–4	Runs the program selected via the CMDS panel PRGM knob once, independent of RWR threat state. After execution, the CMDS returns to standby and resumes threat detection.	2.7.2.2	18, 28
CMDS SEMI	Left	Short	Execute Program 6	Program 6 is dispensed independently of the PRGM knob position. After execution, the CMDS returns to standby and resumes threat detection.	2.7.2.2	19
CMDS SEMI	Down	Short	Give Consent; Dispense One Program	Dispenses one program (1–4) in response to a COUNTER prompt. If the threat persists or a new threat appears, the CMDS issues COUNTER again. Consent state is retained: if the pilot switches to AUTO, the CMDS begins dispensing immediately on the next detected threat.	2.7.2.1	18, 28
CMDS SEMI	Right	Short	Cancel Consent; Return to Standby	Removes SEMI consent and places the CMDS in Standby. COUNTER messages cease.	2.7.2.1	18, 28

1.2.2 ECM

The ECM system operates in two configurations: *external* pods (ALQ-131, ALQ-184), controlled via **CMS Down**, and *internal* IDIAS, controlled via **CMS Left**.

1.2.2.1 External ECM Pod (ALQ-131/ALQ-184)

With the ECM pod power switch in OPR, **CMS Down** grants consent for ECM pod transmission and the ECM Enable light on the left miscellaneous panel illuminates.

The XMIT switch on the ECM control panel sets the ECM operation mode: XMIT 1 (Avionics Priority, AFT antenna only), XMIT 2 (ECM Priority, both FWD and AFT antennas), or XMIT 3 (Active Jam, continuous transmission).

When the RF switch is moved out of NORM, all ECM systems are placed in Standby.

Table 1.4: CMS Actuation with External ECM Pod

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
ECM Pod	Down	Short	Grant ECM Consent	The pod transmits in the operation mode set by the XMIT switch. Transmission continues until CMS Right is pressed or the RF switch is moved out of NORM.	2.7.4.1.1, 2.7.4.2.5	28
ECM Pod	Right	Short	Remove ECM Consent	ECM consent is removed and the pod is placed in Standby. The ECM Enable light extinguishes. Transmission does not resume until the pilot presses CMS Down again.	2.7.4.1.1, 2.7.4.2.5	28

1.2.2.2 Internal ECM (IDIAS)

With the XMTR switch in OPER, **CMS Left** cycles the IDIAS ECM operational mode: Standby (STBY), Avionics Priority (AVNC), and ECM Priority (ECM).

The active mode is displayed on the HUD, FCR, RWR, and TFR pages.

Table 1.5: CMS Actuation with Internal ECM (IDIAS)

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
IDIAS ECM	Left	Short	Cycle ECM Operational Mode	The IDIAS mode advances with each short press: STBY → AVNC (Avionics Priority) → ECM (ECM Priority) → AVNC → ECM.	2.7.4.1.2	28
IDIAS ECM	Right	Short	Remove ECM Consent	IDIAS is set to STBY and consent is removed. AVNC or ECM modes are not restored until the pilot presses CMS Left again.	2.7.4.1.2	28

1.2.3 CMDS and ECM Joint Consent

When equipped with an external ECM pod, **CMS Down** grants consent to *both* CMDS (in AUTO or SEMI mode) and the external ECM pod *simultaneously*.

Table 1.6: CMS Interaction with CMDS and ECM (Consent and Constraints)

Mode	Dir.	Act.	Function	Effect / Nuance	Dash34	Train.
CMDS AUTO SEMI + ECM Pod	Down	Short	Joint Consent (CMDS + ECM)	Consent is granted to <i>both</i> the ALE-47 CMDS (AUTO or SEMI mode) and the external ECM pod <i>simultaneously</i> .	2.7.2.1, 2.7.4.2.5	18, 28

1.2.4 Constraints and Edge Cases

State Tracking: In AUTO and SEMI modes, the CMDS retains consent state when the pilot switches to MAN mode. If the pilot re-engages AUTO without pressing **CMS Down**, the CMDS begins dispensing immediately on the next detected threat.

Bingo Quantity Behavior: If expendables fall to or below the bingo quantity, the CMDS continues dispensing. A LOW audio message is issued when an expendable reaches the bingo quantity and an OUT audio message is issued when an expendable is depleted. Dispensing does not stop automatically. The pilot must press **CMS Right** to inhibit AUTO dispensing, or switch to MAN mode to take direct control.

RF Switch Override: When ECM consent is active, the ECM system is placed in Standby if the RF switch is moved from NORM to QUIET or SILENT. Returning the RF switch to NORM does *not* restore consent. The pilot must press **CMS Down** again to resume ECM transmission.

IDIAS vs. External ECM Pod: The joint consent described in [Section 1.2.3](#) and in [Table 1.6](#) applies to external ECM pod configurations *only*. On IDIAS-equipped aircraft, **CMS Left** cycles ECM modes and **CMS Down** has no ECM function.

Ground Safety: On the ground, ECM consent state is Standby. If the pilot holds **CMS Down** on the ground, radiation is emitted until **CMS Down** is released. If ECM must be enabled on the ground, ground personnel must be clear of the radiation area beforehand.

1.3 ECM Configuration by Block and Variant

CMS interaction with CMDS is *uniform* across all F-16 blocks and variants (see [Section 1.2.1](#)). External Pod and IDIAS F-16 variants, however, use *different* CMS procedures and ECM control panels (see [Section 1.2.2](#)):

Table 1.7: ECM Configuration by Variant — CMS Interface Summary

ECM Configuration	CMS Direction	Panel Control
External Pod (ALQ-131/ALQ-184)	CMS Down (consent)	XMIT switch (positions: 1, 2, 3)
Internal IDIAS	CMS Left (mode cycling)	XMTR switch (positions: STBY, OPER)

1.3.1 ECM Configurations in BMS

Falcon BMS 4.38.1 implements two ECM configurations. Each configuration uses different CMS actuation procedures, described in [Section 1.2](#).

The variants listed in the following tables reflect Dash-34 documentation; not all may be available as selectable aircraft in Falcon BMS 4.38.1.

1.3.1.1 ECM Pods (ALQ-131 / ALQ-184)

These variants use the ECM Pod Control Panel with manual jamming band selection. **CMS Down** grants consent for ECM transmission. For detailed procedures, see [Section 1.2.2.1](#).

Table 1.8: External ECM Pod Variants

Operator	Block/Variant
USAF	Blocks 40/42/50/52 (CM designation)
NATO	Block 15 operators (Belgium, Denmark, Netherlands, Norway)
International	Egypt, Korea KF-16C Block 32

1.3.1.2 Internal ECM (IDIAS)

These variants use the IDIAS ECM control panel. **CMS Left** cycles the operational mode. For detailed procedures, see [Section 1.2.2.2](#).

Table 1.9: Internal ECM (IDIAS) Variants

Operator	Block/Variant
Israel	F-16I Sufa Block 52, Barak I Block 30, Barak II Block 40
International	Greek HAF (Blocks 50 PXII, 52 PXIII, 52+ PXIV Advanced), Korea KF-16C Block 52, Singapore F-16D Block 52